Anurag Yadav

Infrastructure Systems Engineer · Grounded in EPC Execution

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Ithaca, New York

PROFILE

Infrastructure systems engineer with eight years delivering EPC projects across civil, structural, and MEP systems, including multi-project industrial and commercial developments up to 250 acres in scale. Graduate work in systems engineering at Cornell applied MBSE, digital twins, optimization under uncertainty, and verification methods to the coordination, interface, and operational-risk problems that show up in complex infrastructure delivery. Work spans project execution, technical coordination, systems integration, commissioning, and infrastructure modeling across both physical and digital engineering domains.

TECHNICAL FOCUS

Capital Program Delivery · EPC Coordination · Mission-Critical MEP Integration · Systems Engineering · MBSE · Infrastructure Modeling · Verification & Validation · Interface Control · Digital Twins · Optimization Under Uncertainty · Commissioning · Program Controls

SELECTED PROJECTS

Wind Turbine Digital Twin — Cornell · SYSEN 5490

CONTEXT A browser-deployed reduced-order twin spanning aerodynamics, rotor inertia, pitch and yaw control, and electrical generation.

CHALLENGE Calibrating five coupled physical parameters against manufacturer specifications, then verifying behavior across five phases: baseline physics, Cp surface fidelity, turbulence response, transient control, and design optimization.

OUTCOME Peak Cp of 0.48 (81% of the Betz limit), a 19.4% modeled capacity factor, and a documented ~7× gap between the analytical AEP surrogate and the dynamic twin, exactly the kind of finding the verification was built to surface.

Infrastructure Scheduling Under Uncertainty — Cornell · SYSEN 5211

CONTEXT A mixed-integer scheduling model for several concurrent infrastructure projects facing variable, uncertain demand.

CHALLENGE Deciding which operating constraints reflected field reality rather than which were simply easy to quantify, then testing robustness across 10,000 Monte Carlo runs.

OUTCOME Traffic disruption fell about 27% and mean cost dropped about $35K against the deterministic baseline, which consistently underestimated operational risk.

Independent V&V of a Locally Deployed LLM (Gemma 4B) — Cornell · SYSEN 5400

CONTEXT A verification framework for a non-deterministic AI system, run as an independent V&V study.

CHALLENGE 1,250 scored observations across five decoding temperatures, with formal pass/fail criteria and hallucination categories mapped by domain.

OUTCOME 93.6% of test cells stabilized below CV 0.10, and the study surfaced a structural-math failure mode that point-accuracy metrics missed entirely evidence for verifying bounded behavior rather than single-shot accuracy.

MBSE Smart Bus Shelter — Cornell × CUSD / TCAT Capstone

CONTEXT Full systems-engineering lifecycle in SysML and Cameo for a solar-powered transit shelter.

CHALLENGE Catching maintenance access, energy autonomy, emergency-response timing, and stakeholder conflicts early enough to resolve them on paper from stakeholder requirements through FMEA (12 critical failure modes), trade studies, and a V&V plan.

OUTCOME A 2.2 kW solar design with 72-hour autonomy validated against 36 tracked requirements, about 35% lower energy use than a conventional shelter, and a roughly 8-year lifecycle payback.

EXPERIENCE

Project Manager -Capital Program Delivery & Technical Coordination Nov 2021 – Jul 2025

Avalon Developers · Pune, India · 250-acre, six-project capital program

- Brought on-time delivery from 76% to 94% across six concurrent projects on a 250-acre development by establishing dependency tracking, a formal RFI and submittal workflow, and closer coordination among the civil, structural, MEP, and survey teams.
- Identified terrain, alignment, and waterway conflicts through early-stage spatial analysis in Google Earth Pro before formal survey mobilization, preventing over $100K in survey work that would not have been usable.
- Delivered an $8M, 40,000 sq ft facility with a 30-meter clear-span roof, holding schedule through repeated scope changes while coordinating structural, MEP, and architectural interfaces in parallel.
- Served as the engineering point of contact to ownership, finance, and sales, translating design constraints into terms the commercial side could decide on and building the feasibility models that gated whether projects entered detailed design.

Civil Engineer -Infrastructure Integration & Commissioning Nov 2018 – Oct 2021

Nyati Group · Pune, India · $15M, four-project commercial portfolio

- Reduced post-occupancy defects by roughly 31% across a $15M, four-project commercial portfolio by tightening integration testing, commissioning, and LEED-related MEP coordination.
- Cut schedule variability by about 28% through scenario planning and recurring dependency reviews carried across the project lifecycle.
- Ran MEP, structural, and civil interface management from design through commissioning, issuing interface control documents and holding constructability reviews as scope evolved.

Junior Engineer -Structural Systems & Interface Coordination Jul 2016 – Nov 2018

Manthan Construction India Pvt. Ltd. · Pune, India · $8M, five-project industrial portfolio

- Resolved 45+ design conflicts before they reached site across five industrial and commercial projects ($8M, 180,000 sq ft of PEB and RCC work) by coordinating earlier across the design, procurement, and construction handoffs.
- Reduced material waste by about 14% while holding 89% on-schedule delivery through tighter subcontractor sequencing.

EDUCATION

Master of Engineering, Systems Engineering May 2026

Cornell University · Ithaca, NY

Coursework: MBSE · Digital Twins & Smart Infrastructure · Simulation Modeling · System Architecture & Integration · Operations Research · Project Management

Bachelor of Engineering, Civil Engineering May 2016

Savitribai Phule Pune University · Pune, India

CERTIFICATION

Associate Systems Engineering Professional (ASEP) - INCOSE Jan 2026